

Building ML/AI Applications

Introduction and administration

Carlos Cotrini and the teaching team

CAS ETH in AI and Software Development D-INFK, ETH Zürich

Autumn semester 2026



Section 1

About us

About Carlos



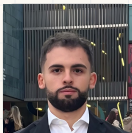
Dr. Carlos Cotrini

- BSc Mathematics, 2009
- MSc Information Security, ETH, 2014
- PhD Information Security, ETH, 2019
- PostDoc in AI and ML, ETH, 2019 to 2021
- Senior scientist, ETH, 2021 to now

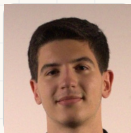
Courses at ETH and UZH

- Advanced machine learning, 2019 to now
- Statistical learning theory, 2020 to 2022
- AI in medicine, 2020 to 2022
- Data analysis in physics, 2020 to 2025
- Stochastics and machine learning, 2024 to now
- Building AI/ML applications, 2024 to now
- AI and IT in industry, 2025 to now
- Software engineering fundamentals, 2025 to now
- AI project, 2026 to now
- Applied introduction to machine learning, from 2027

The teaching team



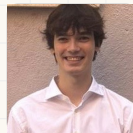
Ivan Angjelovski



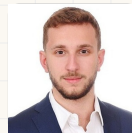
Amine Benjelloun
Touimi



Ghali Berbich



Francesco Caperna



Konstantinos Chasiotis



Suei-Wen Chen



Luca Chimienti



Monica De Alvaro Mena



Mattia de Martino

The teaching team



Vassili De Palma



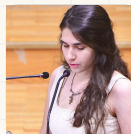
Hassan El Bouz



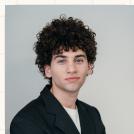
Alper Karadeniz



Joel Kaufmann



Matina Lampaki



Angelo Marano



Samy Messouak



Elisej Orsini-Rosenberg



Frederick Puls

The teaching team



Kangsheng Qi



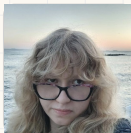
Lamberto Ragnolini



Winston Rohmann



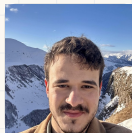
Daniele Russica



Sanziana Stelea



Christina Tsakanika



Celal Turkmen



Jinlei Zhou

We study how we teach this

Transforming Confusion into Diffusion SIGCSE TS 2026

CER Best Paper (Global) award

- **Full stack ML:** you build the model from the ground up before you reach for a library.
- Trial with $N = 208$: that beat the library first approach by about **10 percent** on conceptual understanding ($p = 0.006$), and left participants more curious.

PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning ICER 2026

- “...multi-foundational: they draw on multiple distinct fields and have complex prerequisite structures.”
- The order we teach things in is itself designed. Trial with $N = 209$: the designed order scored higher ($p = 0.011$, $d = 0.32$) at lower cognitive load.

Experimental-study team Dr. Sverrir Thorgeirsson · Stefan Cadar · Sofi Mousia · Umut Demirbas

Johannes Schulte-Vels · Nikolaj Murasov · Jonathan Maillefaud · Oscar Jauffret

Building ML/AI Applications HS26, introduction and administration

Transforming Confusion into Diffusion: Advancing Machine Learning Education via Bottom-Up Instruction

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ACM Reference Format:
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Abstract

Reducing conceptual depth with practical skill development is a perennial challenge in advanced machine learning (ML) education, where powerful frameworks can obscure underlying mathematical and computational principles. To address this, we crafted a new pedagogical approach that we call full-stack machine learning (FSML), which emphasizes the construction of large language models and diffusion models from scratch. To evaluate the effectiveness of FSML, we conducted a classroom-based randomized controlled trial (RCT) in which FSML-based instruction was compared against a popular library-based instructional approach. We measured the study's conceptual understanding through a specialized assessment and administered a survey capturing knowledge gaps, resources, interests, and cognitive load. We found that students who received FSML instruction performed approximately 10% better than control participants in a quiz on transformers and stable diffusion ($p = 0.040$). They also showed increased curiosity and more positive affective responses, suggesting deeper engagement with ML fundamentals. Our findings indicate that our full-stack approach is effective in improving individual learning outcomes, previously including constraints for ML and other advanced computing topics.

1 Introduction

Computer science (CS) education has many dimensions that are relevant to their colleagues from other disciplines, one of which is how to balance high-level conceptual approaches to the subject with foundational skill development. While top-down approaches that involve lectures and assignments take care of the motivation and structural interest of students [1, 10], they may leave students short of practical proficiency when not paired with consistent practice in hands-on labs—a position well known, for example, in foundational library education [15, 16] and mathematics [11, 12]. Consequently, many education face a kind of top-down and bottom-up methods to equilibrate on the strength of both, although the best way to integrate these is competing education remains a topic of ongoing discussion [19].

In machine learning (ML) education, this question is particularly

PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning

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Abstract

Designing an effective machine learning (ML) curriculum is a complex task. It involves balancing conceptual depth with practical skill development, ensuring that students gain a deep understanding of the underlying principles while also acquiring the necessary tools and techniques for real-world applications.

Our goal is to develop a pedagogical method for sequencing instruction in ML that addresses these challenges. We propose a multi-foundational approach that emphasizes the construction of large language models and diffusion models from scratch, while also providing a solid foundation in the underlying mathematical and computational principles.

We conducted a randomized controlled trial (RCT) to evaluate the effectiveness of our method. The results show that students who received our instruction performed significantly better than the control group on a series of tests designed to measure conceptual understanding and practical skill development.

Our findings indicate that our multi-foundational approach is effective in improving individual learning outcomes, previously including constraints for ML and other advanced computing topics.

ACM Reference Format:
Garcia, Diego Rivera; Thorgeirsson, Sverrir; Meier, Dominik; Seligso, Jana; Gissurarson, Leifur. PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning. In: Proceedings of the 2026 Conference on International Computer Science Education (SIGCSE), Article 1. New York, NY, USA: ACM, 2026. <https://doi.org/10.1145/3711111>.

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Section 2

Artificial intelligence, what is it?

Four paradigms in science



Artificial intelligence, a working definition



The design and implementation of systems that perform tasks normally requiring **human intelligence**.

Main approaches

- **Symbolic reasoning:** rules and logic
- **Machine learning:** estimate dependencies from data
- **Deep learning:** machine learning with deep neural networks

Artificial intelligence, a timeline

Foundations



Turing test, 1950



Dartmouth conference, 1956



Rosenblatt's perceptron, 1958

Expert systems, AI winter

LISP, 1958

DENDRAL and MYCIN, 1970s

Statistical machine learning

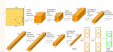


Deep Blue, 1997



IBM Watson, 2011

Deep learning



AlexNet, 2012

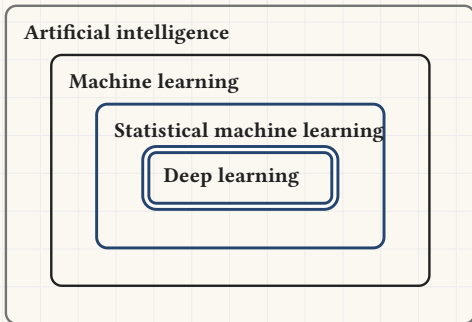


AlphaGo, 2016



ChatGPT, 2022

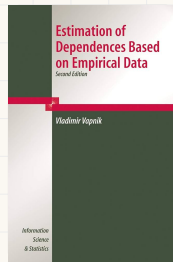
From artificial intelligence to deep learning



- **Artificial intelligence** designs programs that act like intelligent systems, by whichever means.
- **Machine learning** does it by estimating dependencies from observed data.
- **Statistical machine learning** estimates those dependencies with Vapnik's empirical risk minimisation.
- **Deep learning** does the same with deep neural networks.

Vladimir Vapnik, and where this all comes from

- He turned learning from data into a mathematical question.
- His book *Estimation of Dependences Based on Empirical Data* founded statistical machine learning.
- Every model in this course minimises the empirical risk he defined.



Vladimir N. Vapnik, and the 2006 edition

Building ML applications, and hence AI applications.

You learn to use deep learning to design applications that create value from data.

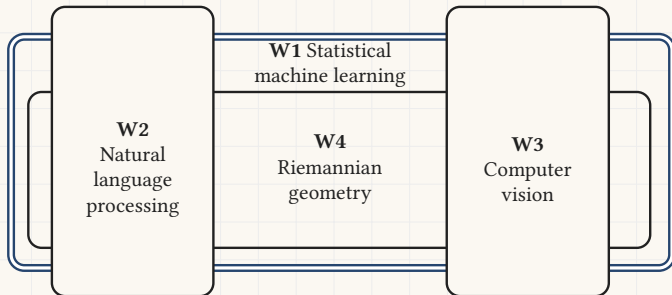
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Section 3

Scope and learning goals

What you learn in this course

This course goes deeper into the machine learning you met earlier in the programme. We explore four areas of modern AI.

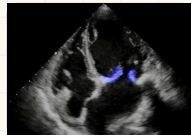


For each area you build an application: a piece of software that uses AI in an innovative way to produce business value.

Some success stories

Oliver Krone, 2025, head of grid and hydro engineering at BKW.

- Entered the Kaggle competition of Advanced machine learning, where students segment the mitral valve of human hearts from echocardiogram videos. Placed 12th out of 100.



The FDD cohort, 2026, a whole class of working professionals.

- Handed the first graded project of *Advanced Machine Learning*, an ETH Master's course, with no scaffolding.
- Their 18 teams beat its 136 Master's teams on mean, median and consistency. Top score: an AML grade of **5.95**.

Bernhard Obrist, 2025, senior project manager at Aargauische Kantonalbank.

- Built an agent that learns on its own to drive a two dimensional road. Carlos's own agent crashes after 15 seconds.
- Demo:
<https://polybox.ethz.ch/index.php/s/raoa3SBoG7cPnT2>



A whole class of professionals, against an ETH Master's course

	AML ($n = 136$)	FDD ($n = 18$)
Mean	0.7115	0.7426
Median	0.7164	0.7420
Std	0.0485	0.0309
Min	0.4865	0.6774
Max	0.8062	0.7922

- Participants of our sister course were handed the first graded project of *Advanced Machine Learning*, an ETH Master's course, with no project structure to lean on.
- Their 18 teams beat its 136 Master's teams on mean, median and consistency. Only **one** AML team beat their best score.
- They did it in **12.6** submissions per team against the Master's students' **25.6**. Their top score converts to an AML grade of **5.95**.

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Section 4

Administration

**Please read these slides
carefully.**

The course website

<https://eth-bmai-hs26.github.io/BMAI-PAGE/>

- The schedule for all four weekends, Friday and Saturday.
- The coding exercises for every session.
- The project descriptions and their grading schemes.



Scan it to open the site on your phone

Passing the course

To pass the course you must do all of the following.

1. Complete the **first three** of the four applications and submit your code on Moodle:
<https://moodle-app2.let.ethz.ch/course/view.php?id=28802>
2. Defend each of those three in a 15 minute presentation to your team's teaching assistant.
3. Pass **three** of the four Moodle quizzes. You may retake a quiz as often as you need, but you must reach at least 80 percent of the points.

Everything above must be finished before the grand deadline

Sunday 25 October 2026, 23:59

The applications

- Each application is tied to one of the four areas above. We release one after each weekend, in order. **Four are offered, three are required.**
- We recommend that you finish each application within one month.
- Applications are solved **in pairs** wherever possible.
- Form your team on Moodle. A teaching assistant is assigned to your team, and **it is the same assistant for all three defences:**
<https://moodle-app2.let.ethz.ch/mod/choicegroup/view.php?id=1434210>
- Arrange each defence directly with that assistant, over Zoom, and hold it before the grand deadline.

Weekend	Application
W1, Vapnik's statistical learning theory	LLM routing
W2, Natural language processing	Tax agent
W3, Computer vision	Fashion magazine editor
W4, Riemannian geometry	Deep fake generation and detection

The four applications

One application per weekend, each one a piece of software that uses AI to produce business value. **The first three are required.**



W1 LLM router

Statistical machine learning

Send each question to the cheapest model that can still answer it well.



W2 Tax agent

Natural language processing

Read tax documents and answer questions about them in plain language.



W3 Fashion magazine editor

Computer vision

Produce the images from text: describe the shot, get the picture.



W4 Deep fake generation and detection

Riemannian geometry

Build the generator, then build the detector that catches it.

Google Colab

- We recommend Google Colab for this course: <https://colab.research.google.com/>
- You get a Google Colab Pro licence, valid until the grand deadline.
- To claim it you first set up a Google workspace. Instructions: unlimited.ethz.ch, Google Workspace, first steps

Office hours

- The teaching team holds office hours over Zoom between the weekends.
- **Tuesday, Wednesday and Thursday, 18:00 to 20:00**, in 30 minute slots.
- **Book a slot on the course website.** Enter your name, then click an open slot. Anyone can cancel a booking.
- Two teaching assistants are on duty each day. The meeting link is the one that assistant posts, and it appears under their name on the same page.

<https://eth-bmai-hs26.github.io/BMAI-PAGE/#/office-hours>



Scan it to book a slot

Questions?



`eth-bmai-hs26.github.io/BMAI-PAGE`